

WENO-Z Reconstruction Equations

Arithmetic extracted from `reconstruct.F90`

For each reconstructed variable, let q_j denote its cell-centered value at index j . The reconstruction uses the five-point stencil $q_{i-2}, \dots, q_{i+2}$ to form states at the two interfaces surrounding cell i .

Candidate interface values

Three third-order candidates are formed for each interface:

$$\begin{aligned} P_1 &= \frac{1}{3}q_{i-2} - \frac{7}{6}q_{i-1} + \frac{11}{6}q_i, \\ P_2 &= -\frac{1}{6}q_{i-1} + \frac{5}{6}q_i + \frac{1}{3}q_{i+1}, \\ P_3 &= \frac{1}{3}q_i + \frac{5}{6}q_{i+1} - \frac{1}{6}q_{i+2}, \\ M_1 &= -\frac{1}{6}q_{i-2} + \frac{5}{6}q_{i-1} + \frac{1}{3}q_i, \\ M_2 &= \frac{1}{3}q_{i-1} + \frac{5}{6}q_i - \frac{1}{6}q_{i+1}, \\ M_3 &= \frac{11}{6}q_i - \frac{7}{6}q_{i+1} + \frac{1}{3}q_{i+2}. \end{aligned}$$

Here P_k contribute to the right-face state u^+ at $i + \frac{1}{2}$; M_k contribute to the left-face state u^- at $i - \frac{1}{2}$.

Smoothness indicators

The same three indicators are used to weight both interfaces:

$$\begin{aligned} \beta_1 &= \frac{13}{12}(q_{i-2} - 2q_{i-1} + q_i)^2 + \frac{1}{4}(q_{i-2} - 4q_{i-1} + 3q_i)^2, \\ \beta_2 &= \frac{13}{12}(q_{i-1} - 2q_i + q_{i+1})^2 + \frac{1}{4}(q_{i-1} - q_{i+1})^2, \\ \beta_3 &= \frac{13}{12}(q_i - 2q_{i+1} + q_{i+2})^2 + \frac{1}{4}(3q_i - 4q_{i+1} + q_{i+2})^2, \\ \tau_5 &= |\beta_1 - \beta_3|. \end{aligned}$$

WENO-Z nonlinear weights

For interface direction $s \in \{+, -\}$ and stencil $k \in \{1, 2, 3\}$,

$$\alpha_k^{(s)} = \gamma_k^{(s)} \left(1 + \frac{\tau_5}{\beta_k + \epsilon} \right), \quad \omega_k^{(s)} = \frac{\alpha_k^{(s)}}{\alpha_1^{(s)} + \alpha_2^{(s)} + \alpha_3^{(s)}},$$

with $\epsilon = 10^{-36}$ and linear weights

$$(\gamma_1^+, \gamma_2^+, \gamma_3^+) = (0.1, 0.6, 0.3), \quad (\gamma_1^-, \gamma_2^-, \gamma_3^-) = (0.3, 0.6, 0.1).$$

Combine, flatten, and limit

First combine the candidates with their direction-specific weights:

$$u_{\text{WENO}}^+ = \sum_{k=1}^3 \omega_k^+ P_k, \quad u_{\text{WENO}}^- = \sum_{k=1}^3 \omega_k^- M_k.$$

Then blend each state toward the cell value using the flattening coefficient $f = \text{flat}$:

$$u^+ = f u_{\text{WENO}}^+ + (1 - f) q_i, \quad u^- = f u_{\text{WENO}}^- + (1 - f) q_i.$$

Finally, if the monotonicity test

$$(u^+ - q_i)(q_i - u^-) \leq 0$$

holds, replace both interface states by the cell value:

$$u^+ = q_i, \quad u^- = q_i.$$